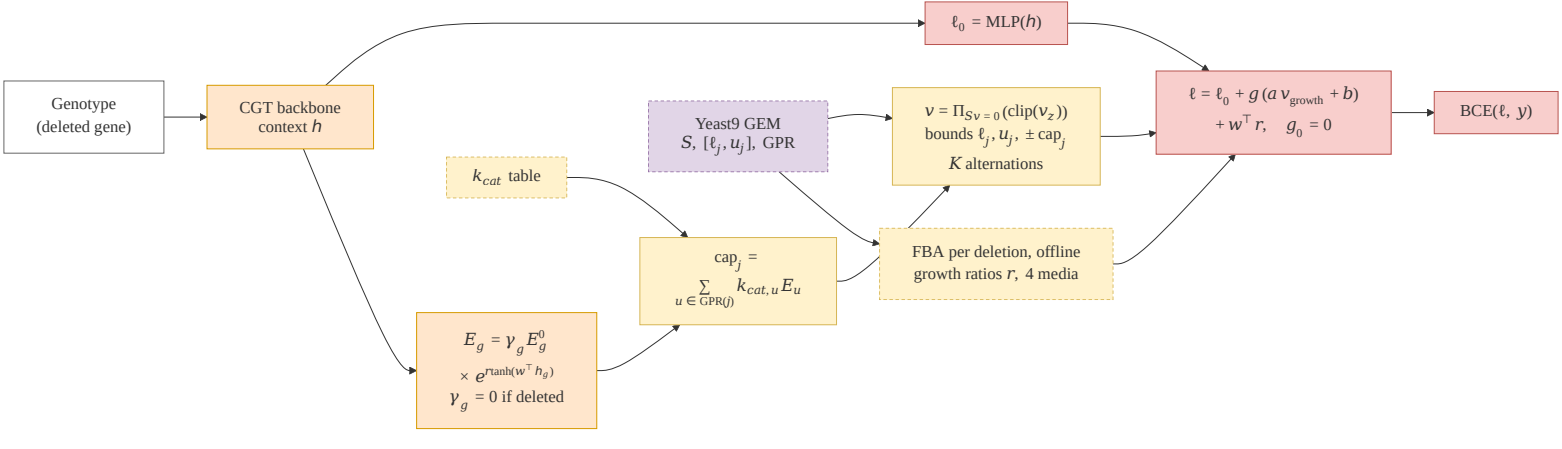

Genotype
(deleted gene)
CGT backbone
context h
$\ell_0 = \mathrm{MLP}(h)$
Yeast9 GEM
$S,\ [\ell_j, u_j],\ \mathrm{GPR}$
k_{cat} table
$\nu = \Pi_{S\nu=0}(\mathrm{clip}(\nu_z))$
bounds $\ell_j, u_j, \pm \mathrm{cap}_j$
K alternations
$\ell = \ell_0 + g\,(a\,\nu_{\mathrm{growth}} + b)$
$+ w^\top r, \quad g_0 = 0$
BCE(ℓ, y)
$\mathrm{cap}_j = \sum_{u \in \mathrm{GPR}(j)} k_{cat,u} E_u$
FBA per deletion, offline
growth ratios r, 4 media
$E_g = \gamma_g E_g^0$
$\times\ e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ if deleted